

HEATHER PRINCE

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EDUCATION

2018-2022	PhD in Astrophysical Sciences in progress Advisor: Professor Jo Dunkley <i>Princeton University, USA</i>
May 2020 - Apr 2021	Maternity leave then primary caregiver for my child due to COVID-19
2016-2018	Master of Arts Awarded on completion of the General Examination Department of Astrophysical Sciences <i>Princeton University, USA</i>
2014-2015	Master of Science received cum laude Thesis: Gravitational Lensing of the Cosmic Microwave Background: Techniques and Applications Advisor: Professor Kavilan Moodley <i>University of KwaZulu-Natal, South Africa</i>
2013	Bachelor of Science Honors received summa cum laude Research Project: Real Space Gravitational Lensing Reconstruction from CMB Temperature and Polarization Advisor: Professor Kavilan Moodley <i>University of KwaZulu-Natal, South Africa</i>
2010-2012	Bachelor of Science with distinction Majors: Mathematics and Physics <i>Rhodes University, South Africa</i>

PUBLICATIONS

H. Prince and J. Dunkley, *Compressed Python likelihood for large scale temperature and polarization from Planck*. Accepted for publication in Phys. Rev. D (2021), [arXiv:2104.05715](#).

H. Prince and J. Dunkley, *Data compression in cosmology: A compressed likelihood for Planck data*. Phys. Rev. D Volume 100 Issue 8 (2019), [arXiv:1909.05869](#).

H. Prince, K. Moodley, J. Ridl., M. Bucher. *Real space lensing reconstruction using cosmic microwave background polarization*. JCAP 01 (2018) 034., [arXiv:1709.02227](#).

Papers as a member of the Atacama Cosmology Telescope collaboration

S. Aiola, E. Calabrese, L. Maurin, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: DR4 maps and cosmological parameters*. JCAP 12 (2020) 047. [arXiv:2007.07288](#).

S. K. Choi, M. Hasselfield, S.-P. P. Ho, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: a measurement of the Cosmic Microwave Background power spectra at 98 and 150 GHz*. JCAP 12 (2020) 045. [arXiv:2007.07289](#).

R. Datta, S. Aiola, S. K. Choi, et al. (including **H. Prince**). *The Atacama Cosmology Telescope: two-season ACTPol extragalactic point sources and their polarization properties*. MNRAS 486, 5239 (2019). [arXiv:1811.01854](#).

Partridge, B., Bonavera, L., López-Caniego, M. et. al. (including **H. Prince**). *Can CMB Surveys Help the AGN Community?* Galaxies, 5, 47 (2017).

SCIENTIFIC PRESENTATIONS

October 2021	Characterizing primordial fluctuations with the Atacama Cosmology Telescope Astronomy Seminar , Columbia University
October 2021	Constraining isocurvature fluctuations with the Atacama Cosmology Telescope and the Simons Observatory Journal Club , Rutgers University
January 2020	Data compression and likelihood-free inference in cosmology American Astronomical Society Meeting , Hawaii
March 2019	Data compression in cosmology: A lightweight <i>Planck</i> likelihood Cosmology on Safari Conference , South Africa
January 2018	Variability of blazars at 148 GHz in the Atacama Cosmology Telescope data Atacama Cosmology Telescope collaboration meeting , Princeton
December 2015	Cross-correlating CMB lensing with post-reionisation HI intensity mapping surveys Square Kilometre Array Bursary Conference , Stellenbosch, South Africa
September 2015	Cross-correlating CMB lensing with post-reionisation HI intensity mapping surveys University of KwaZulu-Natal Research Day , Pietermaritzburg, South Africa Award for runner up to best presentation
January 2015	Real space lensing reconstruction using cosmic microwave background polarization Cosmology on Safari Conference , South Africa
December 2014	Real space lensing reconstruction using cosmic microwave background polarization Square Kilometre Array Bursary Conference , Stellenbosch, South Africa Award for runner up to best MSc Astronomy presentation
November 2014	Real space lensing reconstruction using cosmic microwave background polarization South African Gravity Society Conference , Cape Town, South Africa Award for best postgraduate student oral presentation
November 2014	Real space lensing reconstruction using cosmic microwave background polarization Dark Side of the Universe Conference , Cape Town, South Africa
September 2014	Real space lensing reconstruction using cosmic microwave background polarization University of KwaZulu-Natal Research Day , Pietermaritzburg, South Africa
July 2014	Real space lensing reconstruction using cosmic microwave background polarization South African Institute of Physics Conference , Johannesburg, South Africa Award for best MSc oral presentation in the Astrophysics division

OUTREACH PRESENTATIONS

May 2019	Junior Girl Scout Event , Cranford NJ
April 2019	Girl Scout Career Day , Springfield NJ
2014-2015	School Presentations Talks on astrophysics at several high schools

ORGANIZER

September 2021	Mental health workshop for graduate students , Princeton NJ Invited a representative from Counseling and Psychological Services to run a wellness workshop and share mental health resources for graduate students
December 2019	Department discussion on Maunakea and the Thirty Meter Telescope , Princeton NJ Co-organized a department wide discussion on the conflict surrounding the use of Maunakea for astronomy, including consulting with a lecturer in anthropology who studies colonialism and co-leading a presentation followed by a discussion session
Fall 2018	Peyton Yoga , Princeton NJ Offered free weekly yoga classes to students and staff in the department

RELEVANT COURSEWORK AND WORKSHOPS

Spring 2021	Fundamentals of Machine Learning , Princeton University, USA Applying machine learning techniques to social justice and scientific problems, primarily using scikit-learn, including a final project in which I used Gaussian Process regression to predict supernova distances from features extracted from their lightcurves
Spring 2021	Modern Statistics , Princeton University, USA A model based Bayesian approach to data analysis focused on leveraging probabilistic programming frameworks
July 2019	Workshop on the Boltzmann-Einstein solvers CLASS and SONG , run by Julien Lesgourges and Christian Fidler, Center for Computational Astrophysics, New York, USA
Spring 2017	Physics of the Universe , Princeton University, USA Inflation and the Λ CDM model and issues in contemporary cosmology
Fall 2016	Software Engineering for Scientific Computing , Princeton University, USA Best practices in scientific computing including testing, version control
December 2013 & 2014	Radio Astronomy School , University of KwaZulu-Natal, Durban, South Africa Lectures on radio astronomy theory and techniques, data analysis group project
January 2014	Exploiting Nature's Telescopes: A first look at the Hubble Space Telescope Frontier Fields , Durban, South Africa Gravitational lensing conference and HST data analysis project

TECHNICAL SKILLS

Programming Languages	C/C++, Python
Markup Languages	L ^A T _E X
Software	scikit-learn NumPy, SciPy, pandas Astropy Cosmological parameter estimation codes including CosmoSIS and Cobaya Boltzmann codes including CLASS and CAMB Standalone MCMC sampling codes including emcee Simulation based inference codes including pydelfi

TEACHING AND MENTORING

Fall 2019	Physics Instructor for Prison Teaching Initiative Co-taught an introductory physics course in East Jersey State Prison <i>Princeton University, USA</i>
Spring 2019	Mathematics Instructor for Prison Teaching Initiative Co-taught an introductory mathematics course in Edna Mahan Correctional Facility <i>Princeton University, USA</i>
Fall 2018	Mathematics Teaching Assistant for Prison Teaching Initiative Tutored students for a precalculus course in East Jersey State Prison <i>Princeton University, USA</i>
2018-2019	Undergraduate mentoring Mentored an undergraduate student as part of the Undergraduate Women in Physics Program <i>Princeton University, USA</i>
Fall 2017	Assistant in Instruction for AST205: Planets in the Universe Hold office hours and precepts, set and grade assignments and exams <i>Princeton University, USA</i>
2014-2015	Astrophysics and Cosmology Research Unit (ACRU) Tutor Tutor undergraduate students in physics with a focus on applications to astronomy <i>University of KwaZulu-Natal, South Africa</i>
2012	Physics Laboratory Demonstrator Assist in leading physics and electronics laboratories for first year students <i>Rhodes University, South Africa</i>
2011-2012	Physics Academic Development Program Tutor Additional tutoring for first year physics students <i>Rhodes University, South Africa</i>

AWARDS

Princeton 2016-2021	Dean's Grant
Princeton 2016	Princeton University First Year Fellowship
UKZN 2014	Vincent Maphai Scholarship awarded to the top-ranked Master's student in the university Square Kilometer Array MSc Bursary
Rhodes 2012	Rhodes University Foundation Scholarship awarded to the top student graduating with a bachelor's degree David Williams Memorial Prize for Mathematics awarded to the top third year Mathematics student Alexander Ogg Prize for Physics in recognition of Academic Achievement Academic Honors Investec Rhodes Top 100 Award for Academic Excellence in Science
Rhodes 2011	Rhodes Governors Scholarship awarded to the top second year student in the Faculty of Science or Pharmacy Maryam Babangida Scholarship awarded to the top female second year student Sydney Cruise Memorial Prize awarded to the top second year Mathematics student Trevor Williams Prize awarded to the top second year Physics student Janinne Franke Prize awarded to the top second year Computer Science student

HOBBIES

Yoga (qualified yoga teacher)
Hiking
Skiing
Reading, especially epic fantasy novels

REFERENCES

Jo Dunkley
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Michael Strauss
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Kavilan Moodley
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